



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

NATIONAL SENIOR CERTIFICATE NASIONALE SENIOR SERTIFIKAAT

GRADE 12/GRAAD 12

TECHNICAL MATHEMATICS P2/TEGNIESE WISKUNDE V2

NOVEMBER 2025

MARKING GUIDELINES/FINALE NASIENRIGLYNE

MARKS/PUNTE: 150

CODE/ KODE	EXPLANATION/VERDUIDELIKING
A	Accuracy/Akkuraatheid
AO	Answer only/Slegs antwoord
CA	Consistent accuracy/Volgehoue akkuraatheid
I	Identity/Identiteit
M	Method/Metode
NPR	No penalty for rounding/Geen penalisering vir afronding nie
NPU	No penalty for omitting units/Geen penalisering vir eenhede weggelaat nie
R	Rounding/Afronding
RE	Reason/Rede
S	Simplification/Vereenvoudiging
F	Formula/Formule
SF	Substitution in correct formula/Vervanging in korrekte formule
ST	Statement/Bewering
ST/RE	Statement with reason/Bewering met rede

**These marking guidelines consist of 32 pages.
Hierdie nasienriglyne bestaan uit 32 bladsye.**

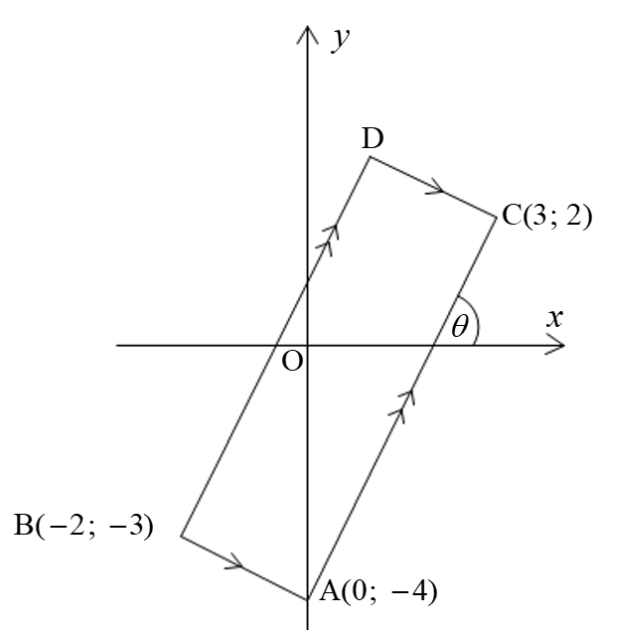
NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- The method of consistent accuracy marking must be applied in all aspects of the marking guidelines where applicable as indicated by the marking code CA.
- # Shows questions where a Tolerance Range will be applied:
Q 2.1.3 ; Q 3.3 ; Q 6.5 & Q 10.2.3

LET WEL:

- Indien 'n kandidaat 'n vraag TWEE keer beantwoord, sien slegs die EERSTE poging na.
- Die metode van volgehoue akkuraatheid-nasien moet waar moontlik op alle aspekte van die nasienriglyne toegepas word, soos aangedui deur die nasien kode CA.
- # Toon vrae waar Toleransiewydte (Verdraagsaamheidsomvang) toegepas word:
V 2.1.3 ; V 3.3 ; V 6.5 & V 10.2.3

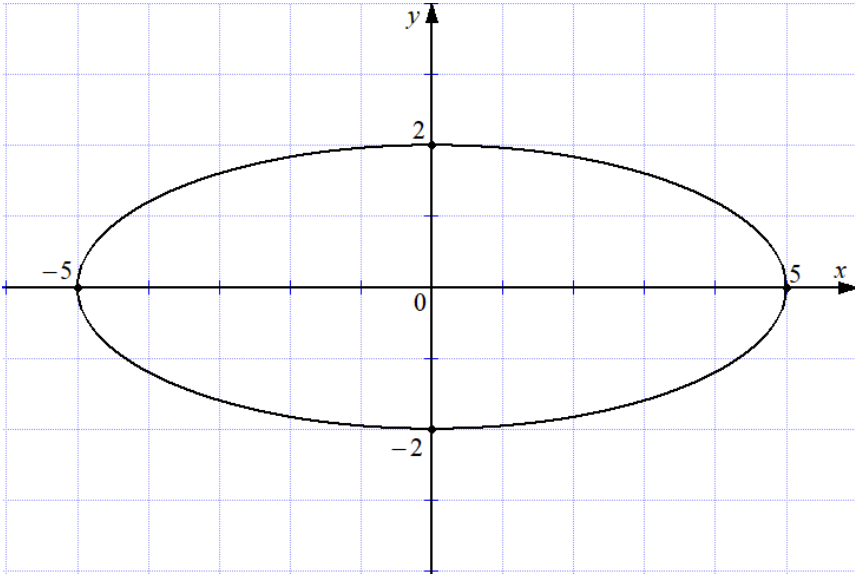
QUESTION/VRAAG 1

			
1.1	OA = 4 units/eenhede	✓ length of /lengte van OA	A (1)
1.2	$M = \left(\frac{x_A + x_B}{2}; \frac{y_A + y_B}{2} \right)$ $= \left(\frac{0 + (-2)}{2}; \frac{-4 + (-3)}{2} \right)$ $= \left(-1; -\frac{7}{2} \right) \text{ OR / OF } (-1; -3,5)$	✓ x - coordinate/koördinaat ✓ y - coordinate/koördinaat	A A (2)

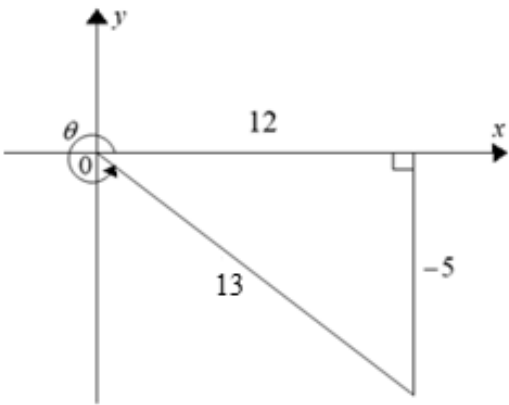
1.3	$m_{AC} = \frac{y_A - y_C}{x_A - x_C}$ $= \frac{(-4) - (2)}{(0) - (3)} \text{ OR / OF } = \frac{(2) - (-4)}{(3) - (0)}$ $= 2$ OR / OF $y - y_1 = m (x - x_1)$ $(-4) - (2) = m ((0) - (3))$ $-6 = m (-3)$ $m = \frac{-6}{-3} = 2$ OR / OF $y = mx + c$ $y = mx - 4$ $2 = m(3) - 4$ $3m = 6$ $m = 2$	✓ SF A ✓ value of/waarde van m CA OR / OF ✓ SF A ✓ value of/waarde van m CA OR / OF ✓ SF A ✓ value of/waarde van m CA AO: Full marks/Volpunte (2)
1.4	$\tan \theta = m_{AC}$ $\theta = \tan^{-1}(2)$ $\theta \approx 63,43^\circ$	✓ SF CA ✓ value of/waarde van θ CA AO: Full marks/Volpunte NPR (2)
1.5	equal / gelyk OR / OF the same / dieselfde	✓ equal/gelyk A (1)
1.6	$y - (-3) = 2 (x - (-2))$ $y + 3 = 2x + 4$ OR / OF $y = 2x + 1$	✓ gradient/helling CA ✓ Subst./Vervang B A ✓ equation in form $y = \dots$ / vergelyking in vorm $y = \dots$ CA (3)

1.7	$\cos \alpha = -\frac{\sqrt{2}}{2}$ ref. / verw. $\angle = 45^\circ$ $\therefore \alpha = 180^\circ - 45^\circ = 135^\circ$ $\therefore m = \tan(135^\circ) = -1$ OR / OF $\cos \alpha = -\frac{\sqrt{2}}{2}$ $x^2 + y^2 = r^2$ $(-\sqrt{2})^2 + y^2 = (2)^2$ $y^2 = 4 - 2$ $y^2 = \pm\sqrt{2}$ $\therefore y = \sqrt{2}$ $\tan \alpha = \frac{\sqrt{2}}{-\sqrt{2}}$ $\tan \alpha = -1$ $\therefore m = -1$	✓ Ref. $\angle$ / Verwys. $\angle$ A ✓ value of/waarde van α CA ✓ value of/waarde van m A OR / OF ✓ Value of/Waarde van y A ✓ tan ratio/verhouding CA ✓ value of/waarde van m CA (3)
		[14]

2.1.3 #	Since / Omdat M(13; y) $m_{ON} = \frac{12}{5}$ $m_{MN} = -\frac{5}{12}$ $y - (-12) = -\frac{5}{12}(x - (-5)) \text{ OR / OF } -12 = -\frac{5}{12}(-5) + c$ $y = -\frac{5}{12}x - \frac{25}{12} - 12 \qquad c = -\frac{25}{12} - 12 = -\frac{169}{12}$ $\therefore y = -\frac{5}{12}x - \frac{169}{12}$ $\therefore y = -\frac{5}{12}(13) - \frac{169}{12}$ $\therefore y = -\frac{39}{2} = -19,5$ $\therefore M(13; -19,5)$ OR / OF Since / Omdat M(13; y) $x \cdot x_1 + y \cdot y_1 = r^2$ $-5x - 12y = 169$ $-12y = 5x + 169$ $\therefore y = -\frac{5}{12}x - \frac{169}{12}$ $\therefore y = -\frac{5}{12}(13) - \frac{169}{12}$ $\therefore y = -\frac{39}{2} = -19,5$ $\therefore M(13; -19,5)$ OR / OF Since / Omdat M(13; y) and/en $d_{LM} = d_{NM}$ $\left(\begin{array}{l} \text{tangents from same pt. /} \\ \text{Raaklyne vanuit dies pt} \end{array} \right)$ $\sqrt{(13 - (-5))^2 + (y - (-12))^2} = y$ $(13 - (-5))^2 + (y - (-12))^2 = y^2$ $324 + y^2 + 24y + 144 = y^2$ $24y = -468$ $y = -\frac{39}{2} = -19,5$ $M(13; -19,5)$	✓ value of/waarde van x A ✓ gradient/gradiënt of/van ON CA ✓ gradient/gradiënt of/van MN CA ✓ equation/vergelyking CA ✓ value of/waarde van y CA OR / OF ✓ value of/waarde van x A ✓ F A ✓ subst/vervang CA ✓ equation/vergelyking CA ✓ value of/waarde van y CA OR / OF ✓ value of/waarde van x A ✓ statement/bewering A ✓ subst/vervang CA ✓ squaring both sides / kwadreer beide kante CA ✓ value of/waarde van y CA (5)
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2.2		✓ x-intercept/afsnit A ✓ y-intercept/afsnit A ✓ elliptical shape/ elliptiese vorm A (3) ACCEPT/AANVAAR: $\sqrt{25}$ and / en $-\sqrt{25}$ $\sqrt{4}$ and / en $-\sqrt{4}$
		[12]

QUESTION/VRAAG 3

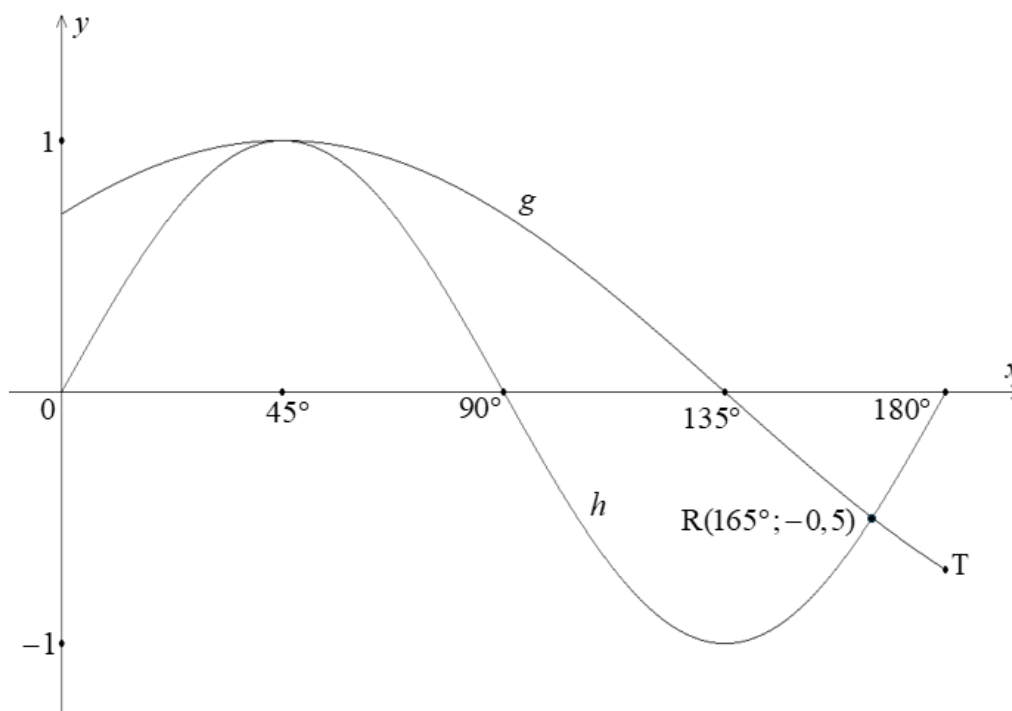
3.1	$\sqrt{\sin B + \sec A}$ $= \sqrt{\sin 30,5^\circ + \sec 72^\circ}$ $= \sqrt{\sin 30,5^\circ + \frac{1}{\cos 72^\circ}}$ $\approx 1,93$	✓ substitution/vervanging A ✓ I A ✓ S CA NPR AO: Full marks/Volpunte (3)
3.2.1	 $\sin \theta = -\frac{5}{13}$ $x^2 + (-5)^2 = (13)^2$ $x = 12$ $\cos \theta = \frac{12}{13}$ OR / OF $\cos \theta = \sqrt{1 - \sin^2 \theta}$ $= \sqrt{1 - \left(-\frac{5}{13}\right)^2}$ $= \frac{12}{13}$	✓ Pyth. Theorem/Stelling A ✓ value of/waarde van x CA ✓ ratio/verhouding CA OR / OF ✓ I A ✓ ratio of/verhouding van sin A ✓ ratio/verhouding CA (3)
3.2.2	$\cot \theta - \operatorname{cosec} \theta$ $= \left(-\frac{12}{5}\right) - \left(-\frac{13}{5}\right)$ $= \frac{1}{5}$	✓ cot ratio/verhouding CA ✓ cosec ratio/verhouding A ✓ S CA ACCEPT/AANVAAR: Correct decimal values./ Korrekte desimale waardes. (3)

3.3 #	$\cot x = -0,587$ $\tan x = -\frac{1}{0,587}$ OR / OF $\tan x = -1,704$ Ref. $\angle \approx 59,59^\circ$ $x \approx 180^\circ - 59,59^\circ$ or/of $x \approx 360^\circ - 59,59^\circ$ $x \approx 120,41^\circ$ or/of $x \approx 300,41^\circ$	✓ tan ratio/verhouding A ✓ ref. angle/verw. hoek CA ✓ value of/waarde van x CA ✓ value of waarde van x CA (4)
		[13]

QUESTION/VRAAG 4

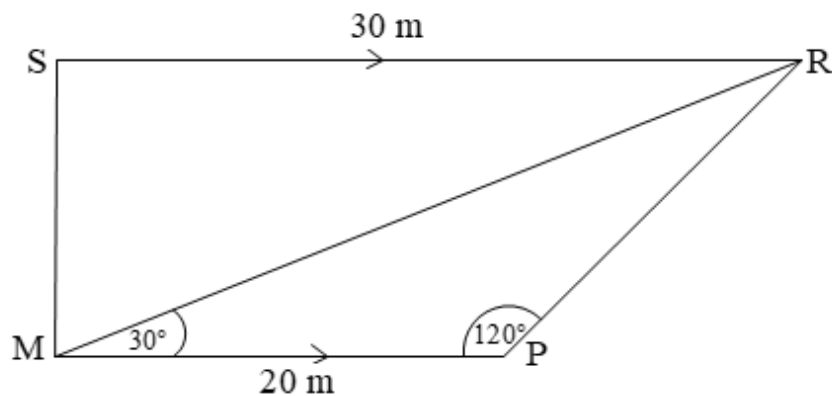
4.1.1	$\tan x$	✓ reduction/ reduksie A (1)
4.1.2	$\frac{\cos x}{\sin x}$	✓ I A (1)
4.1.3	$\{ 0^\circ ; 90^\circ ; 180^\circ ; 270^\circ ; 360^\circ \}$	✓✓ any two values/ enige twee waardes A (2)
4.1.4	$\frac{\sin(180^\circ + x) \cdot \sin(360^\circ - x) + \cos(2\pi - x) \cdot \cos x}{\sin x} + \frac{1}{\tan(180^\circ + x)}$ $= \frac{-\sin x \cdot -\sin x + \cos x \cdot \cos x}{\sin x} + \frac{1}{\tan x}$ $= \frac{\sin^2 x + \cos^2 x}{\sin x} + \frac{\cos x}{\sin x}$ $= \frac{1}{\sin x} + \frac{\cos x}{\sin x}$ $= \frac{1 + \cos x}{\sin x} \text{ OR / OF } = \operatorname{cosec} x + \cot x$	✓ $-\sin x$ A ✓ $-\sin x$ A ✓ $\cos x$ A ✓ I (1) A ✓ S CA (5)
4.2.1	$\sin \theta (1 - \cos \theta)$	✓ factors/ faktore A (1)

4.2.2	$\begin{aligned} \text{LHS} &= \frac{\sin \theta - \cos \theta \cdot \sin \theta}{\cos \theta - (1 - \sin^2 \theta)} \\ &= \frac{\sin \theta(1 - \cos \theta)}{\cos \theta - \cos^2 \theta} \\ &= \frac{\sin \theta(1 - \cos \theta)}{\cos \theta(1 - \cos \theta)} \\ &= \frac{\sin \theta}{\cos \theta} \\ &= \tan \theta = \text{RHS} \end{aligned}$ OR / OF $\begin{aligned} \text{LHS} &= \frac{\sin \theta - \cos \theta \cdot \sin \theta}{\cos \theta - (1 - \sin^2 \theta)} \\ &= \frac{\sin \theta - \cos \theta \cdot \sin \theta}{\cos \theta - 1 + \sin^2 \theta} \\ &= \frac{\sin \theta(1 - \cos \theta)}{\cos \theta - 1 + 1 - \cos^2 \theta} \\ &= \frac{\sin \theta(1 - \cos \theta)}{\cos \theta(1 - \cos \theta)} \\ &= \frac{\sin \theta}{\cos \theta} \\ &= \tan \theta = \text{RHS} \end{aligned}$	✓ I (cos² θ) A ✓ common fact/ gemeensk. fakt A ✓ S A OR / OF ✓ I (1 – cos² θ) A ✓ common fact/ gemeensk. fakt A ✓ S A (3)
		[13]

QUESTION/VRAAG 5

5.1	$m = 2$ and / en $p = 45^\circ$	✓ value of/waarde van p A ✓ value of/waarde van m A (2)
5.2	180°	✓ period/periode A (1)
5.3	1	✓ value/waarde A (1)
5.4.1	$165^\circ < x \leq 180^\circ$ OR / OF $x \in (165^\circ ; 180^\circ]$ OR / OF $x > 165^\circ$ and / en $x \leq 180^\circ$ OR / OF Greater than 165° and smaller or equal to 180° . / Groter as 165° en kleiner of gelyk aan 180° .	✓ endpoints/eindpunte A ✓ notation/notasie A OR / OF ✓ endpoints/eindpunte A ✓ notation/notasie A OR / OF ✓ endpoints/eindpunte A ✓ notation/notasie A OR / OF ✓ endpoints/eindpunte A ✓ Correct interval/Korrekte interval A (2)

5.4.2	$0^\circ \leq x \leq 90^\circ$ $135^\circ \leq x \leq 180^\circ$ OR / OF $x \in [0^\circ ; 90^\circ]$ or / of $x \in [135^\circ ; 180^\circ]$ OR / OF $x \geq 0^\circ$ and / en $x \leq 90^\circ$ or / of $x \geq 135^\circ$ and / en $x \leq 180^\circ$ OR / OF From 0° to 90° / Vanaf 0° tot by 90° or / of From 135° to 180° / Vanaf 135° tot by 180°	✓ critical values/ kritiese waardes A ✓ notation/ notasie A ✓ critical values/ kritiese waardes A ✓ notation/ notasie A OR / OF ✓ critical values/ kritiese waardes A ✓ notation/ notasie A ✓ critical values/ kritiese waardes A ✓ notation/ notasie A OR / OF ✓ critical values/ kritiese waardes A ✓ notation/ notasie A ✓ critical values/ kritiese waardes A ✓ notation/ notasie A OR / OF ✓ endpoints/ eindpunte A ✓ Correct interval/ Regte interval A ✓ endpoints/ eindpunte A ✓ Correct interval/ Regte interval A (4)
5.5	$h(x) = \sin 2x - 1$ OR / OF $h(x) = \sin mx - 1$	✓ $\sin 2x - 1$ CA OR / OF ✓ $\sin mx - 1$ A (1)
		[11]

QUESTION/VRAAG 6**NOTE:**

If an angle of 90° is assumed – **BREAKDOWN**, hence no marks may be awarded.

NOTA:

Indien 'n hoek van 90° aangeneem word = **AFBREEK**, vervolgens mag geen punte toegeken word nie.

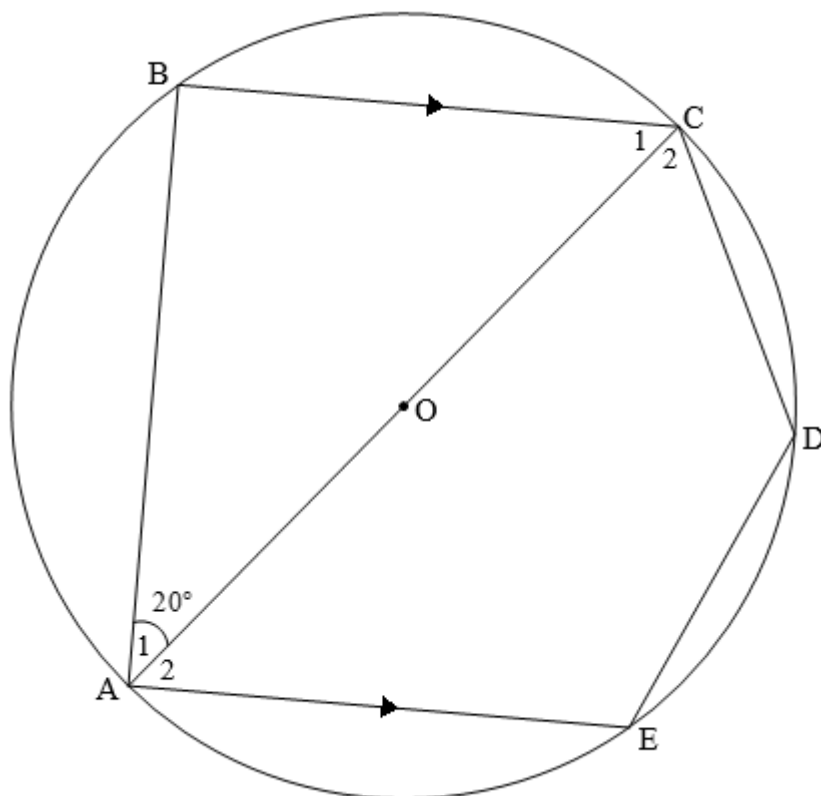
6.1	$\hat{M} \hat{R} \hat{P} = 180^\circ - 120^\circ - 30^\circ$ $= 30^\circ$	✓ size of/grootte van $\hat{M} \hat{R} \hat{P}$ A (1)
6.2	Isosceles triangle/Gelykbenige driehoek OR / OF Obtuse angled triangle/Stomphoekige driehoek	✓ isosceles/gelykbenig A OR / OF ✓ obtuse angled/Stomphoekig A (1)
6.3	$\frac{MR}{\sin P} = \frac{MP}{\sin \hat{M} \hat{R} \hat{P}}$ $\frac{MR}{\sin 120^\circ} = \frac{20}{\sin 30^\circ}$ $MR = \frac{20 \sin 120^\circ}{\sin 30^\circ}$ $MR = 20\sqrt{3} \text{ units} \approx 34,64 \text{ units}$ OR / OF $MP = PR = 20 \text{ m}$ $MR = \sqrt{(20)^2 + (20)^2 - 2(20)(20)\cos 120^\circ}$ $= 20\sqrt{3} \text{ units} \approx 34,64 \text{ units}$	✓ sine rule/sinusreël A ✓ Subst. / Vervang. CA ✓ length of/lengte van MR CA OR / OF ✓ length / lengte PR A ✓ Subst. into / Vervang. in cosine rule/cosinusreël CA ✓ length /lengte MR CA (3)

6.4	$\hat{MRS} = 30^\circ$ (alt. $\angle$ s/verw. $\angle$ e; $SR \parallel MP$)	✓ ST A ✓ RE A (2)
6.5 #	$SM^2 = SR^2 + MR^2 - 2SR \cdot MR \cos \hat{MRS}$ $SM^2 = (30)^2 + (20\sqrt{3})^2 - 2(30)(20\sqrt{3})\cos 30^\circ$ $= 300$ $SM = 10\sqrt{3} \approx 17,32$ $SM^2 + SR^2 = (10\sqrt{3})^2 + (30)^2$ $= 1200$ $MR^2 = (20\sqrt{3})^2$ $= 1\,200$ $\therefore SM^2 + SR^2 = MR^2$ $\therefore \triangle MRS$ is a right-angled triangle $\therefore \triangle MRS$ is 'n reghoekige driehoek OR / OF $SM^2 = SR^2 + MR^2 - 2SR \cdot MR \cos \hat{MRS}$ $SM^2 = (30)^2 + (20\sqrt{3})^2 - 2(30)(20\sqrt{3})\cos 30^\circ$ $= 300$ $SM = 10\sqrt{3} \approx 17,32$ $\cos \hat{S} = \frac{(10\sqrt{3})^2 + (30)^2 - (20\sqrt{3})^2}{2(10\sqrt{3})(30)}$ $\hat{S} = \cos^{-1}\left(\frac{0}{600\sqrt{3}}\right)$ $\hat{S} = 90^\circ$ $\therefore \triangle MRS$ is a right-angled triangle $\therefore \triangle MRS$ is 'n reghoekige driehoek OR / OF	✓ SF cosine rule/ <i>cosinusreël</i> CA ✓ length of/ <i>lengte van</i> SM CA ✓ value of/ <i>waarde van</i> $SM^2 + SR^2$ CA ✓ value of/ <i>waarde van</i> MR^2 CA ✓ $SM^2 + SR^2 = MR^2$ CA OR / OF ✓ SF cosine rule/ <i>cosinusreël</i> CA ✓ length of/ <i>lengte van</i> SM CA ✓ SF cosine rule/ <i>cosinusreël</i> CA ✓ $\cos^{-1}\left(\frac{0}{600\sqrt{3}}\right)$ CA ✓ Size of/ <i>grootte van</i> $\hat{S}$ CA OR / OF

$SM^2 = SR^2 + MR^2 - 2SR \cdot MR \cos \hat{MRS}$ $SM^2 = (30)^2 + (20\sqrt{3})^2 - 2(30)(20\sqrt{3})\cos 30^\circ$ $= 300$ $SM = 10\sqrt{3} \approx 17,32$ $\frac{\sin \hat{S}}{s} = \frac{\sin \hat{R}}{r}$ $\frac{\sin \hat{S}}{20\sqrt{3}} = \frac{\sin 30^\circ}{10\sqrt{3}}$ $\sin \hat{S} = 1$ $\hat{S} = \sin^{-1}(1)$ $\hat{S} = 90^\circ$ $\therefore \Delta MRS$ is a right-angled triangle $\therefore \Delta MRS$ is 'n reghoekige driehoek OR / OF $\sin 30^\circ = \frac{h}{20\sqrt{3}}$ $h = 20\sqrt{3} \cdot \sin 30^\circ$ $h = 10\sqrt{3} \approx 17,32$ $SM = \sqrt{(30)^2 + (20\sqrt{3})^2 - 2(30)(20\sqrt{3}) \cdot \cos 30^\circ}$ $SM = 10\sqrt{3} \approx 17,32$ $\therefore SM$ = Perp. height/<i>loodregte hoogte</i> Therefor SMR is a right-angled triangle. / <i>Daarom is SMR 'n reghoekige driehoek.</i>	✓ SF cosine rule/ <i>cosinusreël</i> CA ✓ length of/<i>lengte van</i> SM CA ✓ SF sine rule/<i>sinusreël</i> CA ✓ $\sin^{-1}(1)$ CA ✓ Size of/<i>grootte</i> van $\hat{S}$ CA OR / OF ✓ sine ratio/<i>verh.</i> CA ✓ h CA ✓ SF cosine rule/ <i>cosinusreël</i> CA ✓ length of/<i>lengte van</i> SM CA ✓ Conclusion/ <i>Gevolgtrekking</i> CA (5)
	[12]

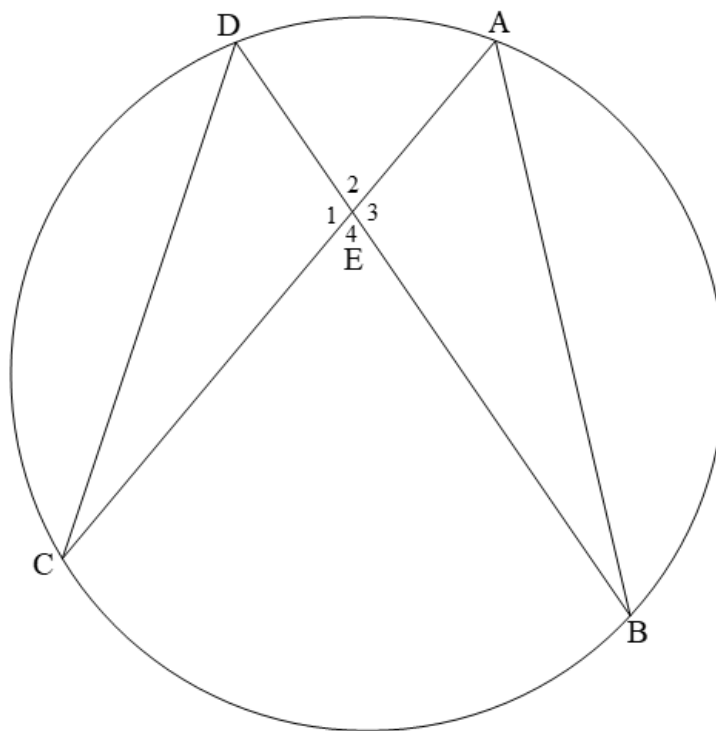
QUESTION/VRAAG 7

7.1



7.1.1	The angle subtended by the diameter at the circumference of the circle is 90°. ($\angle$ in semi-circle) / <i>Die omtrekshoek wat deur die middellyn onderspan word, is 90°. ($\angle$ in halwe sirkel)</i> OR / OF Diameter subtends right angle. / <i>Diameter onderspan regtehoek.</i>	✓ RE A OR / OF ✓ RE A (1)
7.1.2	$\hat{A}_2 = 70^\circ$ (co-int $\angle$s / ko-binne $\angle$e; $BC \parallel AE$) $\therefore \hat{D} = 110^\circ$ (opp $\angle$s of cyclic quad / teenoorst $\angle$e van kdvk) OR / OF $\hat{C}_1 = 70^\circ$ (Int. $\angle$s Δ / Binne $\angle$e Δ; $BC \parallel AE$) $\hat{A}_2 = 70^\circ$ (alt. $\angle$s / verw. $\angle$e; $BC \parallel AE$) $\therefore \hat{D} = 110^\circ$ (opp $\angle$s of cyclic quad / teenoorst $\angle$e van kdvk)	✓ ST A ✓ ST CA ✓ RE A OR / OF ✓ ST A ✓ ST CA ✓ RE A (3)

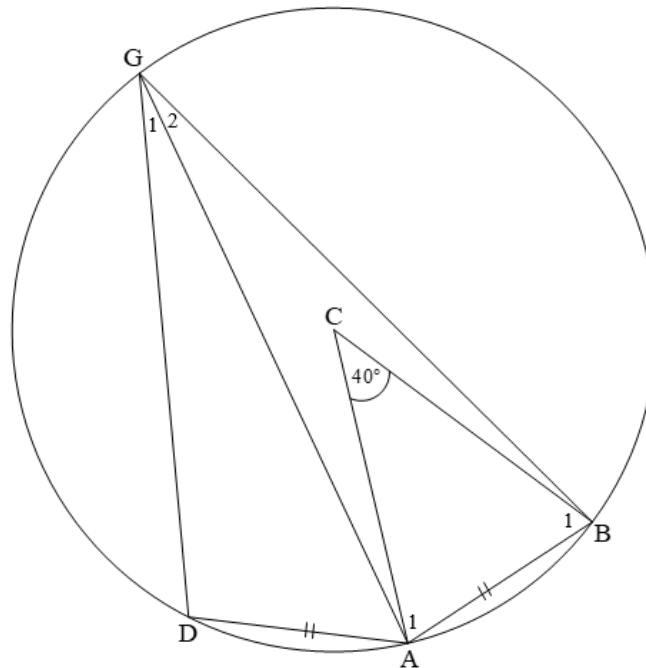
7.2



7.2.1	equal./gelyk. OR / OF the same./dieselfde.	✓ ST	A (1)
7.2.2 (a)	$\angle$ s in the same segment/ $\angle$ e in dieselfde segment OR / OF $\angle$ s subt. by same arc/chord./ $\angle$ e onderspan deur dieselfde boog/koord.	✓ RE OR / OF ✓ RE	A A (1)
7.2.2 (b)	Vertical opposite $\angle$ s / regoorstaande $\angle$ e	✓ ST	A (1)
7.2.2 (c)	$\angle$, $\angle$, $\angle$ OR / OF A, A, A / H, H, H OR / OF Equiangular triangles/Gelykhoekige driehoeke	✓ RE OR / OF ✓ RE	A A (1)
			[8]

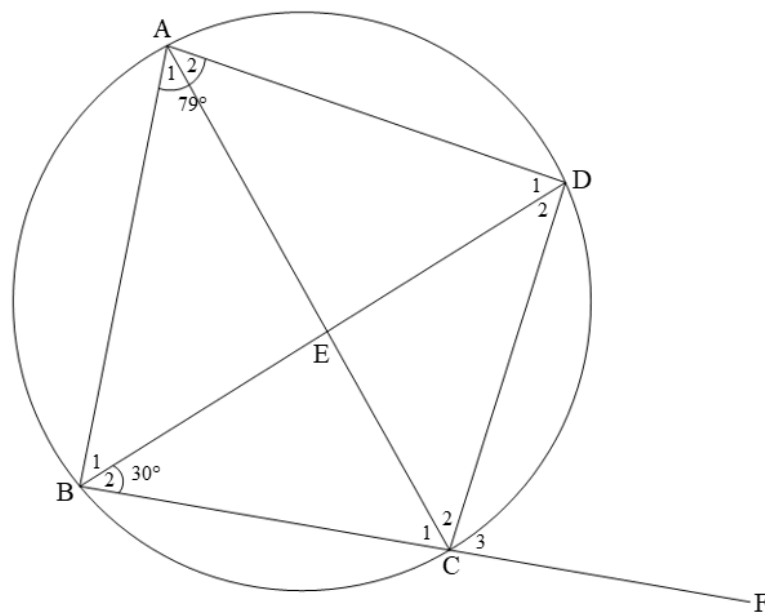
QUESTION/VRAAG 8

8.1



8.1.1	$\hat{G}_1 = \hat{G}_2$	✓ ST	A (1)
8.1.2	$\hat{G}_2 = 20^\circ \left(\begin{array}{l} \angle \text{ at centre} = 2 \times \angle \text{ at circ. /} \\ \text{Mdpt. } \angle = 2 \times \text{omtreks } \angle \end{array} \right)$	✓ ST ✓ RE	A A (2)
8.1.3	$\angle$ s opp = sides / $\angle$ e teenoor gelyke sye	✓ RE	A (1)
8.1.4	$\hat{A}_1 + \hat{B}_1 + \hat{ACB} = 180^\circ \quad (\text{Int. } \angle \text{s } \Delta / \text{Binne } \angle \text{e } \Delta)$ $\hat{A}_1 = \frac{180^\circ - 40^\circ}{2}$ $\hat{A}_1 = 70^\circ$ OR / OF $(\angle \text{s opp.} = \text{sides} / \angle \text{e teenoor} = \text{sye})$	✓ RE ✓ 70°	A A (2)
8.1.5	$\hat{DAC} + 70^\circ + 40^\circ = 180^\circ \quad \left(\begin{array}{l} \text{opp. } \angle \text{s cyclic quad /} \\ \text{teenoorst. } \angle \text{e kdvh} \end{array} \right)$ $\therefore \hat{DAC} = 70^\circ$	✓ ST ✓ RE ✓ ST	CA A CA (3)
8.2.1	exterior / buite	✓ ST ACCEPT/AANVAAR: outside/buitekant	A (1)
8.2.2	interior / binne	✓ ST ACCEPT/AANVAAR: inside/binnekant	A (1)

8.3

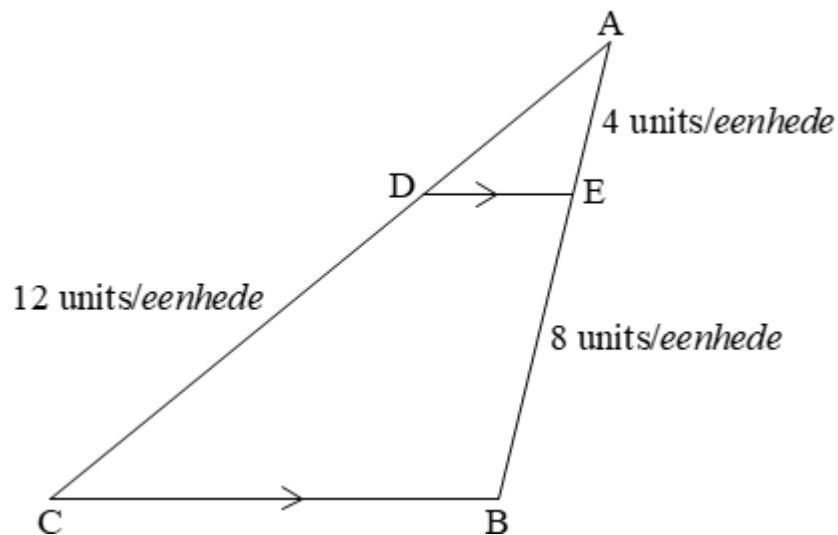


8.3.1(a)	$\hat{C}_3 = 79^\circ$ (Ext. $\angle$ cyclic. quad / <i>Buite $\angle$ kdvh.</i>)	✓ ST ✓ RE	A A (2)
8.3.1(b)	$\hat{D}_2 = 49^\circ$ (Ext. $\angle \Delta$ / <i>Buite $\angle \Delta$</i>) OR / OF $\hat{BCD} = 101^\circ$ ($\angle$ s on str. line / <i>$\angle$ opreguitlyn</i>) $\hat{D}_2 = 49^\circ$ (Int. $\angle$ s Δ / <i>Binne $\angle$ e Δ</i>) OR / OF $\hat{B}_2 = \hat{A}_2 = 30^\circ$ ($\angle$ s in same segm./ $\angle$ e in dies. segm.) $\therefore \hat{A}_1 = 49^\circ$ $\hat{A}_1 = \hat{D}_2 = 49^\circ$ ($\angle$ s in same segm./ $\angle$ e in dies. segm.)	✓ ST ✓ RE OR / OF ✓ ST ✓ ST OR / OF ✓ ST	CA A CA CA CA (2)
8.3.1(c)	$\hat{D}_1 = 180^\circ - 79^\circ - 49^\circ$ (Co - int. $\angle$ s suppl. ; $AB \parallel CD$) $\hat{D}_1 = 52^\circ$ (<i>Ko - Binne $\angle$ e sup pl. ; $AB \parallel CD$</i>) OR / OF $\hat{B}_1 = 49^\circ$ (alt. $\angle$'s =; verw. $\angle$ 'e =; $AB \parallel CD$) $\therefore \hat{D}_1 = 52^\circ$ (Int. $\angle$'s Δ / <i>Binne $\angle$'e Δ</i>)	✓ ST ✓ RE ✓ ST OR / OF ✓ ST ✓ RE ✓ ST	CA A CA CA A CA (3)

8.3.2	$\hat{C}_1 = \hat{D}_1 = 52^\circ \quad \left(\begin{array}{l} \angle s \text{ in same segment /} \\ \angle e \text{ in dies. segment} \end{array} \right)$ $\hat{C}_2 = 49^\circ \quad \left(\begin{array}{l} \angle s \text{ on str. line /} \\ \angle e \text{ op reguitlyn} \end{array} \right)$ $\hat{D}_2 = \hat{C}_2 = 49^\circ \quad \left(\begin{array}{l} \text{sides opp.} = \angle s / \\ \text{syeteenoor} = \angle e \end{array} \right)$ $CE = DE$ OR / OF $\hat{C}_2 = \hat{B}_1 = 49^\circ \quad \left(\begin{array}{l} \angle s \text{ in same segment /} \\ \angle e \text{ in dies. segment} \end{array} \right)$ $\hat{D}_2 = \hat{C}_2 = 49^\circ \quad \left(\begin{array}{l} \text{sides opp.} = \angle s / \\ \text{syeteenoor} = \angle e \end{array} \right)$ $CE = DE$	✓ ST ✓ RE CA A ✓ ST ✓ RE CA A OR / OF ✓ ST ✓ RE CA A ✓ ST ✓ RE CA A (4)
		[22]

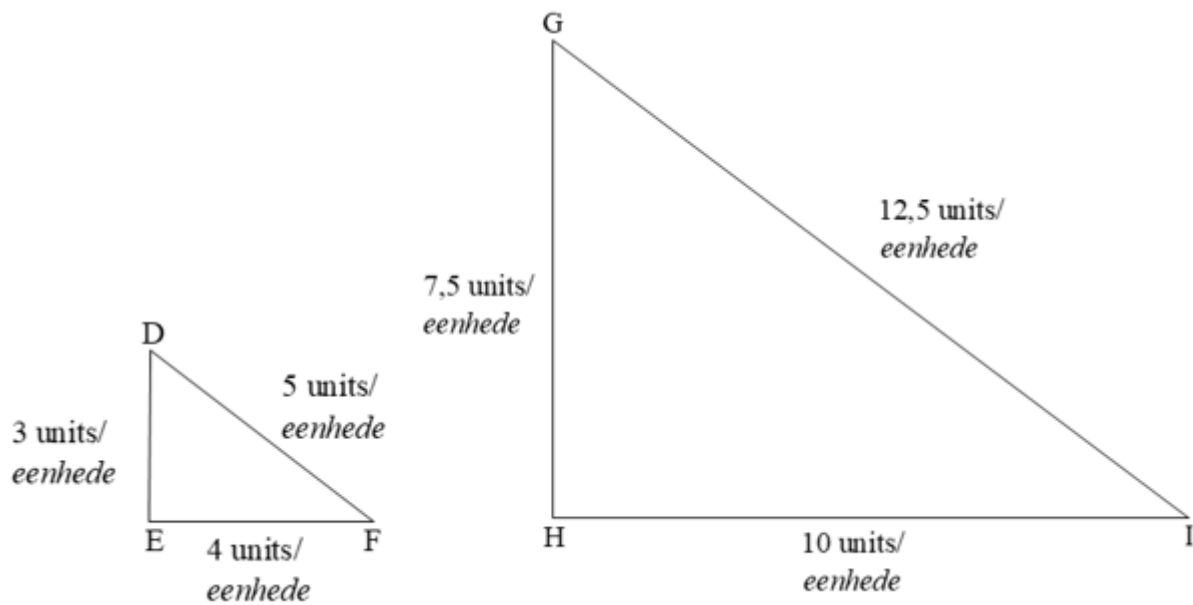
QUESTION/VRAAG 9

9.1



9.1.1	$\frac{AD}{CD} = \frac{AE}{BE}$ (Prop. Theorem / Eweredigheid; $DE \parallel BC$)	✓ ST ✓ RE	A A (2)
9.1.2	$\frac{AD}{CD} = \frac{AE}{BE}$ $\frac{AD}{12} = \frac{4}{8}$ $AD = \frac{4}{8} \times 12$ $= 6 \text{ units / eenhede}$	✓ SF ✓ 6 AO: Full marks/ Volpunte	CA CA (2)

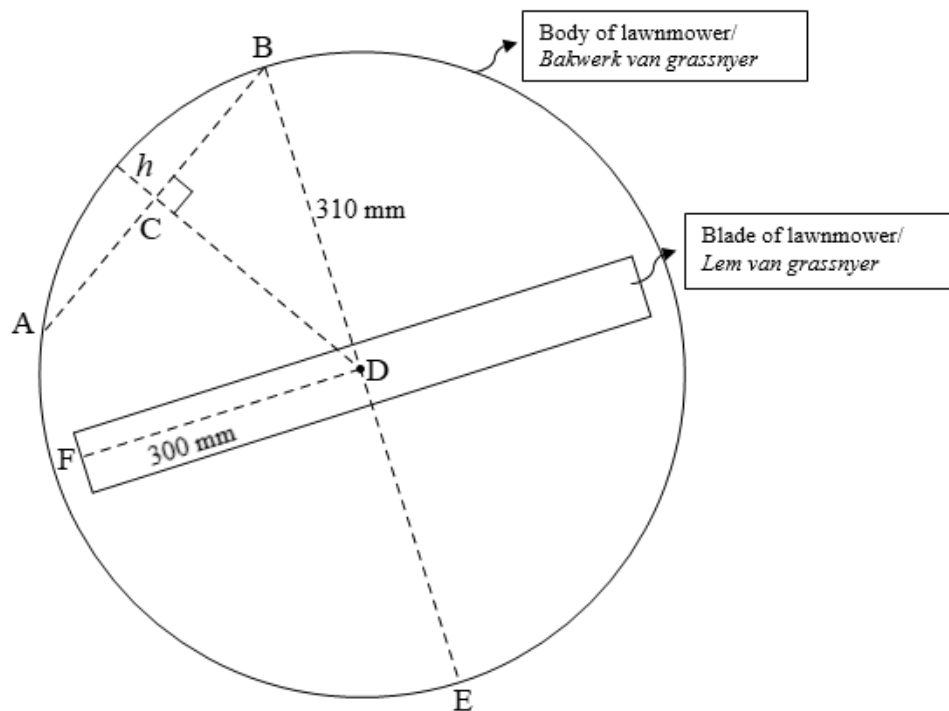
9.2



$\frac{DE}{GH} = \frac{3}{7,5} = \frac{2}{5} = 0,4 \text{ OR / OF } \frac{GH}{DE} = \frac{7,5}{3} = \frac{5}{2} = 2,5$	✓ Ratio/verhouding A
$\frac{EF}{HI} = \frac{4}{10} = \frac{2}{5} = 0,4 \text{ OR / OF } \frac{HI}{EF} = \frac{10}{4} = \frac{5}{2} = 2,5$	✓ Ratio/verhouding A
$\frac{DF}{GI} = \frac{5}{12,5} = \frac{2}{5} = 0,4 \text{ OR / OF } \frac{GI}{DF} = \frac{12,5}{5} = \frac{5}{2} = 2,5$	✓ Ratio/verhouding A
$\therefore \triangle DEF \parallel \triangle GHI \quad (\text{Sides in prop. / Sye in verhouding})$	✓ RE A
OR / OF $\hat{E} = 90^\circ \quad \text{Conv. Pyth. / Omg. Pyth.}$ $\hat{H} = 90^\circ \quad \text{Conv. Pyth. / Omg. Pyth.}$ $\tan \hat{F} = \frac{3}{4} \quad \tan \hat{I} = \frac{3}{4}$ $\hat{F} = 36,87^\circ \quad \hat{I} = 36,87^\circ$ $\hat{E} = \hat{H} \quad (\text{Proven / Bewys})$ $\hat{F} = \hat{I} \quad (\text{Proven / Bewys})$ $\triangle DEF \parallel \triangle GHI \quad (\angle; \angle; \angle)$	OR / OF ✓ Conv. Pyth. / Omg. Pyth. A ✓ Any trig ratio/ Enige trig verhouding A ✓ ST A ✓ RE A (4)
	[8]

QUESTION/VRAAG 10

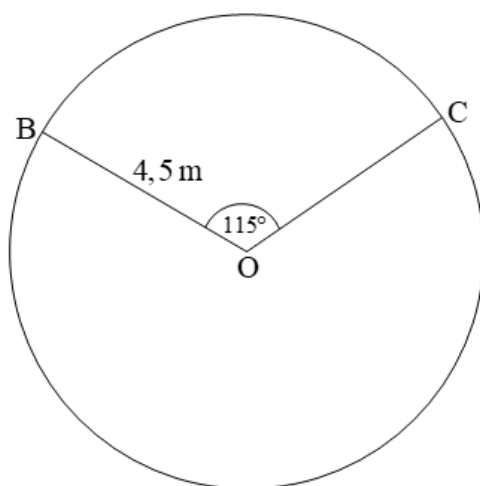
10.1



10.1.1	$\omega = 2\pi n$ $377 = 2\pi n$ $n = \frac{377}{2\pi}$ or / of 60 rev / s	✓ F A ✓ SF A ✓ $\frac{377}{2\pi}$ or / of 60 CA NPU NPR (3)
10.1.2	$v = \omega r$ $= (377)(300)$ $= 113100 \text{ mm / s}$ OR / OF $v = \pi Dn$ $= \pi (600) \left(\frac{377}{2\pi} \right)$ OR / OF $\approx \pi (600)(60)$ $= 113100 \text{ mm / s}$ $\approx 113097,34 \text{ mm / s}$	✓ F A ✓ SF A ✓ value of/waarde van v CA OR / OF ✓ F A ✓ SF CA ✓ value of/waarde van v CA NPU NPR (3)

10.1.3	$4h^2 - 4dh + x^2 = 0$ $4h^2 - 4(620)h + (130)^2 = 0$ $4h^2 - 2480h + 16900 = 0$ $h = \frac{2480 \pm \sqrt{5880000}}{8}$ $h \approx 613,11\text{mm}$ or / of $h \approx 6,89\text{mm}$ $\therefore$ minor / klein $h \approx 6,89\text{mm}$ OR / OF $AC = BC = 65$ $\left(\begin{array}{l} \text{Line from centre of circle, perp. to chord} \\ \text{Lyn mdpt. sirkel, loodreg op koord} \end{array} \right)$ $CD^2 = 310^2 - 65^2$ $CD = 303,11\text{mm}$ $h = 310 - 303,11$ $h = 6,89\text{mm}$	$\checkmark$ F A $\checkmark$ SF A $\checkmark$ S CA $\checkmark$ both vertical heights/beide vertikale hoogtes CA $\checkmark$ minor height / klein hoogte CA OR / OF $\checkmark$ $BC = 65$ A $\checkmark$ M A $\checkmark$ Length/lengte CD CA $\checkmark$ M A $\checkmark$ minor height / klein hoogte CA NPU NPR (5)
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10.2

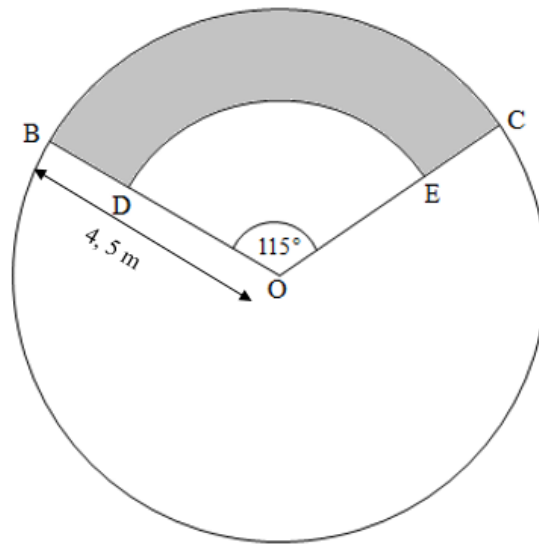


10.2.1	$\hat{BOC} = 115^\circ \times \frac{\pi}{180^\circ}$ $= \frac{23}{36}\pi \approx 2,01 \text{ rad}$	$\checkmark \frac{23}{36}\pi \text{ or/of } 2,01 \quad \mathbf{A}$ (1) NPU NPR
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10.2.2	$\text{Area} = \frac{r^2 \theta}{2}$ $= \frac{(4,5)^2 \left(\frac{23}{36} \pi \right)}{2} \quad \text{OR / OF} \quad \frac{(4,5)^2 (2,01)}{2}$ $= \frac{207}{32} \pi \approx 20,32 \text{ m}^2 \quad \approx 20,35 \text{ m}^2$ OR/OF $\text{Area} = \frac{115^\circ}{360^\circ} \times \pi r^2$ $= \frac{115^\circ}{360^\circ} \times \pi (4,5)^2$ $= \frac{207}{32} \pi \approx 20,32 \text{ m}^2$ OR/OF $s = r \theta$ $= 4,5 \left(\frac{23}{36} \pi \right) \quad \text{OR / OF} \quad = 4,5 (2,01)$ $= \frac{23}{8} \pi \approx 9,03 \text{ m} \quad \approx 9,05 \text{ m}$ $\text{Area} = \frac{r s}{2}$ $= \frac{4,5 \left(\frac{23}{8} \pi \right)}{2} \quad \text{OR / OF} \quad \frac{4,5 (9,05)}{2}$ $= \frac{207}{32} \pi \approx 20,32 \text{ m}^2 \quad \approx 20,36 \text{ m}^2$	✓ F A ✓ SF CA ✓ area/oppervlak CA OR/OF ✓ F A ✓ SF A ✓ area/oppervlak CA OR/OF ✓ arc length/booglengte CA ✓ F A ✓ area/oppervlak CA NPR NPU (3)
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10.2.3

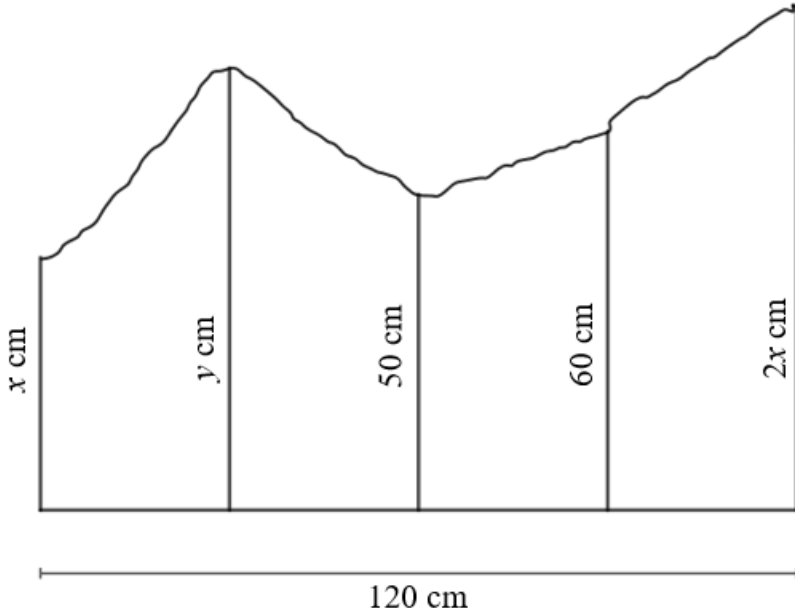
#

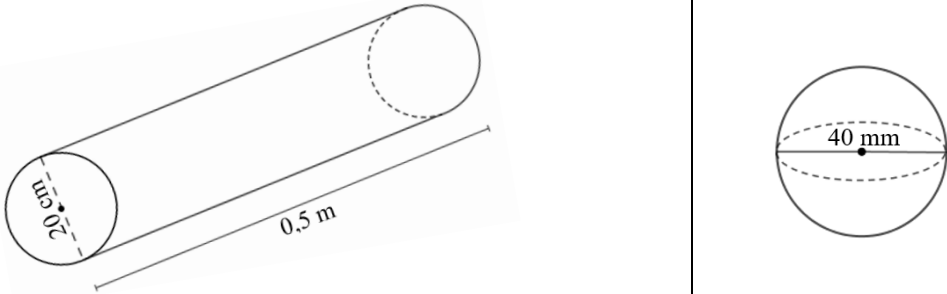


	$DO = \frac{3}{5} \times 4,5 = 2,7 \text{ m}$ $A_{DOE} = \frac{(2,7)^2 \left(\frac{23}{36} \pi \right)}{2} \quad \text{OR / OF} \quad A_{DOE} = \frac{115^\circ}{360^\circ} \times \pi (2,7)^2$ $= \frac{1863}{800} \pi \approx 7,32 \text{ m}^2$ $A_{\text{shaded/geskakeer}} = \frac{207}{32} \pi - \frac{1863}{800} \pi$ $= \frac{207}{50} \pi \approx 13,01 \text{ m}^2$ $\therefore$ Seedlings needed / Saailinge benodig $= 13 \times 4$ $= 52$ OR/OF	✓ ratio/verhouding A ✓ length of/<u>lengte</u> van DO CA ✓ Area of/Oppv. van DOE CA ✓ Area of the shaded part/ Oppv. van geskakeerde gedeelte CA ✓ Number of seedlings/Aantal saailinge CA OR/OF
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	$DO = \frac{3}{5} \times 4,5 = 2,7 \text{ m}$ $s_{DE} = r \theta$ $= 2,7 \left(\frac{23}{36} \pi \right)$ $= \frac{69}{40} \pi \approx 5,42 \text{ m}$ $\text{Area}_{DOE} = \frac{r s}{2}$ $= \frac{2,7 \left(\frac{69}{40} \pi \right)}{2}$ $= \frac{1863}{800} \pi \approx 7,32 \text{ m}^2$ $A_{\text{shaded/geskakeer}} = \frac{207}{32} \pi - \frac{1863}{800} \pi$ $= \frac{207}{50} \pi \approx 13,01 \text{ m}^2$ $\therefore$ Seedlings needed / Saailinge benodig $= 13 \times 4$ $= 52$	✓ ratio/verhouding A ✓ length of/<u>lengte</u> van DO CA ✓ Area of/<i>Oppv. van</i> DOE CA ✓ Area of the shaded part/<i>Oppv. van geskakeerde gedeelte</i> CA ✓ Number of seedlings/<i>Aantal saailinge</i> CA NPU NPR (5)
		[20]

QUESTION/VRAAG 11

11.1			
11.1.1	30 cm	✓ 30 cm	A (1)
11.1.2	$y = 50 + 20 = 70 \text{ cm}$	✓ value of/waarde van y	A (1)
11.1.3	$A_T = a \left(\frac{o_1 + o_n}{2} + o_2 + o_3 + \dots + o_{n-1} \right)$ $7200 = 30 \left(\frac{x + 2x}{2} + 70 + 50 + 60 \right)$ $240 = 1,5x + 180$ $\therefore x = 40 \text{ cm}$ OR / OF $A_T = a (m_1 + m_2 + m_3 + \dots + m_n)$ $7200 = 30 \left(\frac{x + 70}{2} + \frac{70 + 50}{2} + \frac{50 + 60}{2} + \frac{60 + 2x}{2} \right)$ $240 = 1,5x + 180$ $\therefore x = 40 \text{ cm}$		
	✓ F		A
	✓ SF		CA
	✓ S		CA
	✓ value of /waarde van x		CA
	OR / OF		
	✓ F		A
	✓ SF		CA
	✓ S		CA
	✓ value of /waarde van x		CA (4)

11.2		
11.2.1	$0,5\text{ m} \times 100 = 50\text{ cm}$	✓ 50 cm A (1)
11.2.2	$r = 40 \div 2 = 20\text{ mm}$ $\therefore r = 20 \div 10 = 2\text{ cm}$ OR / OF $d = 40 \div 10 = 4\text{ cm}$ $\therefore r = 4 \div 2 = 2\text{ cm}$	✓ M $40 \div 2$ A ✓ r value in/waarde in cm CA OR / OF ✓ M A ✓ r value/waarde in cm CA AO: Full marks/Volpunte (2)
11.2.3	TSA of Cylindrical rod / Buite-oppervlakte van silindriese staaf: $= 2\pi r^2 + 2\pi rh$ $= 2\pi(10)^2 + 2\pi(10)(50)$ $= 1200\pi \approx 3769,91\text{ cm}^2$	✓ SF CA ✓ total surface area/totale buite-oppervlakte CA (2)

11.2.4	Volume of one steel ball bearing/<i>van een staal bal</i> $= \frac{4}{3} \pi (2)^3$ $= \frac{32}{3} \pi \approx 33,51 \text{ cm}^3$ Volume of Cylindrical rod/<i>van silindriese staaf</i> $= \pi (10)^2 (50)$ $= 5000 \pi \approx 15707,96 \text{ cm}^3$ Volume Recovered / <i>Herwin</i> $= 5000\pi - 18\% \times 5000\pi \quad \text{OR / OF}$ $= 82\% \times 5000\pi$ $= 4100 \pi$ $\approx 12880,53 \text{ cm}^3$ $\therefore \frac{4100 \pi}{\frac{32}{3} \pi} \qquad \therefore \frac{32}{3} \pi \times 400$ $= 384,375 \quad \text{OR / OF} \qquad = \frac{12800}{3} \pi$ $\approx 384 \qquad = 13404,13 \text{ cm}^3$ $\therefore$ More than 400 balls can NOT be made / <i>Meer as 400 balle kan NIE gemaak word NIE.</i>	✓ SF CA ✓ Vol. of ball bearing/<i>Vol. van staal koellaër</i> CA ✓ Vol. of cyl. rod/<i>Vol. van sil.staaf</i> CA ✓ M A ✓ Vol. recovered/<i>herwin</i> CA ✓ conclusion/<i>gevolgtrekking</i> CA (6)
		[17]

TOTAL/TOTAAL: 150